01. Gravitation

1. State Kepler's laws of planetary motion. Write any one law in words and explain its meaning in brief.

2. State Newton's universal law of gravitation. Write its mathematical expression and explain the meaning of each symbol used.

3. Using Newton's law of gravitation, derive the inverse square law of gravitation (relation between force and distance).

4. Derive the expression for centripetal force acting on a body moving in a circular path.

5. Derive the expression for acceleration due to gravity (g) on the surface of the Earth.

6. Explain why the value of g varies with altitude. Write the conclusion for g at higher altitude.

7. Explain why the value of g varies with depth. Write the conclusion for g at the centre of the Earth.

8. If the mass of Earth becomes double and its radius becomes half, calculate the new value of g in terms of original g. Justify mathematically.

9. Differentiate between mass and weight on the basis of definition, SI unit and dependence on gravity.

10. Explain why weight of an object changes from place to place but its mass remains constant.

11. Define free fall. State any two characteristics of a freely falling body.

12. A stone is dropped from a height of 125 m. Calculate (a) time taken to reach the ground and (b) velocity with which it strikes the ground. ($g = 10\ m/s^2$)

13. A body is thrown vertically upward with initial velocity 20 m/s. Calculate (a) maximum height reached and (b) time taken to reach the highest point. ($g = 10\ m/s^2$)

14. Two masses of 5 kg and 10 kg are kept 2 m apart. Calculate the gravitational force between them. ($G = 6.67 \times 10^{-11}\ N\ m^2/kg^2$)

15. Define escape velocity. Derive its expression and state its significance in space travel.